CS-4063: Natural Language Processing - Assignment 2

Neural NLP Pipeline on BBC Urdu Corpus

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Overview

This report summarises results for Assignment 2, which constructs a three-part Neural NLP pipeline on a BBC Urdu corpus of 300 articles (~3.5 MB, 10,001-token vocabulary). All components were built from scratch in PyTorch with no pretrained models, HuggingFace, or Gensim. Part 1 builds static word representations (TF-IDF, PPMI, Skip-gram Word2Vec). Part 2 implements a 2-layer BiLSTM with CRF decoding for POS tagging and NER. Part 3 implements a full Transformer encoder for 5-class topic classification. All required output files were saved (embeddings, models, CoNLL data files).

Part 1 Results: Word Embeddings

1.1 TF-IDF and PPMI

A 10,001 × 300 TF-IDF matrix was built from cleaned.txt. Top discriminative tokens per topic were: **Politics**: عمران (Imran), تحریک (Movement), جماعت (Party), قانون (Law); **International**: ایران (Iran), روس (Russia), اسرائیل (Israel), طالبان (Taliban); **Economy**: ٹیکس (Tax), سٹاک (Stock), سرمایہ (Capital), مارکیٹ (Market); **Health & Society**: سیلاب (Flood), ہسپتال (Hospital), ڈاکٹر (Doctor), بارش (Rain).

A 5,000 × 5,000 PPMI co-occurrence matrix (window k = 5, 702,985 non-zero entries) produced semantically valid nearest neighbours: **عدالت (court)** retrieved سماعت (hearing), جج (judge), مقدم (case), وکیل (lawyer); **کرکٹ (cricket)** retrieved کھیل (sport), سٹیڈیم (stadium), ٹیم (team), کھلاڑ (player). A t-SNE visualisation of 200 tokens, colour-coded by topic, was produced with a legend showing clear semantic clustering

2.1 Skip-gram Word2Vec

Skip-gram was trained on cleaned.txt for 20 epochs (d = 100, k = 5, K = 10 negatives, Adam η = 0.001, cosine LR schedule, batch 512, 2,258,106 training pairs after stopword filtering). Nearest neighbors were semantically coherent: **پاکستان (Pakistan)** → انڈا (India), امریکہ (USA), چین (China), افغانستان (Afghanistan); **عدالت (court)** → جج (judge), سماعت (hearing), کورٹ (court), وکیل (lawyer). Final embeddings (V + U)/2 saved as embeddings_w2v.npy.

2.2 Analogy Tests and Four-Condition Comparison

Analogies evaluated with 3CosMul (Levy & Goldberg, 2014) on 10 corpus-validated quadruples (all query words: freq > 60, sentence co-occurrence > 10). **6/10 analogies correct**, including institution–rank (جج:عدالت :: جنرل:فوج), force–country (افغان:طالبان :: پاکستا:فوج), and rank–institution reverse (:: جنرل:فوج عدالت:جج). Failed analogies included افواج:فوج :: سماعت:عدالت and سماعت:عدالت :: حفظان:صحت, where the semantic relationships were not consistently encoded.

Condition	Description	MRR@20
C1 (PPMI)	PPMI co-occurrence (d≈5,000)	0.2492
C2 (raw)	Skip-gram on raw.txt (d=100)	0.2004
C3 (clean)	Skip-gram on cleaned.txt (d=100)	0.1898
C4 (d=200)	Skip-gram on cleaned.txt (d=200)	0.0899

Table 1: MRR@20 across four conditions. PPMI ranks highest on lexical evaluation pairs.

PPMI outperforms Word2Vec on MRR because the evaluation pairs are lexically close neighbors in the raw co-occurrence space. Word2Vec generalizes better to syntactic context but underperforms on direct lexical retrieval with this corpus size. Doubling d to 200 (C4) significantly hurt MRR, suggesting the corpus is too small to fill the higher-dimensional space reliably.

Part 2 Results: Sequence Labelling: POS Tagging & NER Dataset

500 sentences stratified by topic were extracted and annotated (Politics 156, International 170, Health & Society 151, Sports 15, Economy 8). POS lexicon: 205 NOUN, 186 VERB, 116 ADJ, 37 ADV, 22 PRON, 24 DET, 25 CONJ, 30 POST, 37 NUM, 11 PUNC entries. NER gazetteer: 54 people, 78 locations, 44 organizations. Split: Train 347 / Val 73 / Test 80..

5.1 POS Tagging

Mode	Token Accuracy	Macro-F1
Fine-tuned embeddings	0.9893	0.9691
Frozen embeddings	0.8995	0.7308

Table 2: POS tagging test-set results — fine-tuned vs. frozen embeddings.

Fine-tuned embeddings achieved 98.9% token accuracy and 0.969 macro-F1. Closed-class tags scored perfectly (PRON, POST, CONJ, NUM, PUNC all F1 = 1.00). Most confused pairs: NOUN↔UNK (9 errors, e.g., 'جماعت' ,'صدر'), VERB↔UNK (3 errors, e.g., 'ہے'), and NUM↔UNK (3 errors, e.g., 'ایک').

5.2 NER

Model	Precision	Recall	F1
BiLSTM + CRF	0.940	0.922	0.931
BiLSTM (no CRF)	0.811	0.962	0.845

Table 3: Entity-level NER results. CRF decoding yields higher precision and overall F1.

Per-type entity F1 scores for CRF: PER: 0.500, LOC: 0.963, ORG: 0.875, MISC: 0.000. The CRF's transition matrix prevents illegal label sequences (e.g., I-PER following B-LOC), improving person and organisation precision. The no-CRF model had higher recall but lower precision, over-predicting entities. False positives included 'پاس' (tagged PER), 'عادل' (tagged ORG), and 'کے' (tagged PER). False negative included 'ہو کی' (LOC), 'مارچ کے' (LOC), and 'پاس عادل' (ORG).

5.3 Ablation Study (Sequence Labeller)

ID	Ablation	Val F1	Δ vs Full
Full	BiLSTM + CRF (reference)	0.96	—
A1	Unidirectional LSTM only	0.2075	−0.7615
A2	No dropout (p = 0)	0.5589	−0.4101
A3	Random embedding init	0.8454	−0.1637
A4	Softmax output (NER)	0.7599	−0.2091

Table 4: Ablation results on validation set. A1 is catastrophic; A3 confirms embedding value.

A1: Removing the backward pass collapses F1 to 0.208, confirming bidirectionality is essential for Urdu (verb-final, post-positional syntax where POS of early tokens depends on later context). **A2:** No dropout gives a 0.410 F1 drop, indicating overfitting on the small 350-sentence training set. **A3:** Random embeddings cost 0.164 F1, demonstrating the value of Part 1 Word2Vec warm-start. **A4:** Removing the CRF for NER costs 0.209 F1, validating structured decoding for entity recognition.

Part 3 Results: Transformer Encoder for Topic Classification

Setup

Topic labels were assigned using expanded Urdu keywords with title tokens weighted 5×. Distribution after filtering ambiguous articles (MIN_MARGIN = 2): International 71, Politics 67, Health & Society 50, Economy 16, Sports 9. Training set oversampled to 80 per class (Train 400 / Val 30 / Test 36). The Transformer was built from scratch: 4 Pre-LN encoder blocks, 4-head attention (d_model=128), FFN (d_ff=512), sinusoidal PE, learned [CLS] token, MLP head 128→64→5. Total: 2,080,133 parameters. Training: AdamW, $\eta=3\times10^{-4}$, cosine warmup (50 steps), label smoothing 0.1, gradient clipping, early stopping patience 25.

8.1 Results and Comparison

Model	Test Accuracy	Test Macro-F1
BiLSTM (attn+mean+max pool)	0.6389	0.4925
Transformer (from scratch)	0.6111	0.4032
Δ (Transformer − BiLSTM)	−0.0278	

Table 5: 5-class topic classification test-set results.

The BiLSTM (with attention + mean + max pooling, hidden=256) achieved 63.9% accuracy and 0.493 macro-F1, outperforming the Transformer (61.1% / 0.403). The Transformer's per-class breakdown showed reasonable performance on majority classes but struggled with minority classes (Sports, Economy). The core issue is data volume: learning meaningful query-key attention interactions requires thousands of examples, while the BiLSTM's sequential inductive bias generalises well on 300 articles. The BiLSTM also converged in fewer epochs (42 vs. 100). Attention heatmaps revealed that [CLS] tokens attended broadly across input positions, but discriminative patterns were not strongly learned. For datasets of 200–300 articles, the BiLSTM remains the more appropriate architecture.

Conclusion

A complete Neural NLP pipeline was successfully implemented from scratch in PyTorch for a BBC Urdu corpus. Static embeddings (PPMI, Skip-gram) captured meaningful Urdu semantics with PPMI achieving the highest MRR (0.249) on lexical retrieval. The BiLSTM achieved strong sequence labelling results (POS: 98.9% accuracy, NER: 0.931 entity F1) with ablations confirming the value of bidirectionality and pre-trained embeddings. For topic classification on this limited corpus (213 usable articles after filtering), the BiLSTM (63.9% accuracy) outperformed the Transformer (61.1%). Future work could apply cross-lingual transfer, data augmentation, or pretrained multilingual models (e.g., mBERT) to improve Transformer performance on this low-resource task.